



Ecological risk assessment of microplastics in coastal, shelf, and deep sea waters with a consideration of environmentally relevant size and shape[☆]



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ABSTRACT

This study assessed the ecological risk posed by microplastics in surface and subsurface seawaters in coastal, continental shelf, and deep-sea areas of South Korea. The target microplastics for risk assessment were specified as only non-spherical type microplastics in the size range 20–300 μm , because this type was predominantly observed in our study areas, and adverse biological effects have previously been reported. Exposure data for non-spherical microplastics were obtained from a previous study or were measured for microplastics of sizes down to 20 μm . A predicted no-effect concentration (PNEC) of 12 particles/L was derived by employing a species sensitivity distribution approach. Then the results were compared to the *in situ* observed concentrations at each site. The detected microplastic concentrations did not exceed the derived PNEC, i.e., the current pollution levels of fragment and fiber microplastics in the size range 20–300 μm would not pose a significant threat to the marine ecosystem in South Korea. However, predictions are that microplastic pollution will increase to 50-fold by 2100 at the current rates, and in this scenario, the microplastic concentration is expected to far exceed the derived PNEC values for marine ecosystems. It is therefore urgent to take precautionary actions to prevent a further increase in microplastic concentrations in these environments.

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1. Introduction

Pollution by microplastics in aquatic ecosystems represents a significant global issue due to their worldwide detection (Burns and Boxall, 2018; Ivar do Sul and Costa, 2014) and their negative effects on aquatic organisms such as algae, plants, mollusks, crustaceans, echinoderms, and fishes (Au et al., 2015; Cole et al., 2013, 2015; Green, 2016; Imhof et al., 2017; Jeong et al., 2016; Kalčíková et al., 2017; Kaposi et al., 2014; Martínez-Gómez et al., 2017; Mazurais et al., 2015; Ribeiro et al., 2017; Rist et al., 2016; Rochman et al.,

2017; Van Cauwenberghe et al., 2015; Vroom et al., 2017; Ziajahromi et al., 2017). Although researchers and the media have warned of the human and ecosystem health risks of microplastics, the actual risks remain highly uncertain (Koelmans et al., 2017). There has been growing concern over the large mismatch in the sizes and shapes of targeted microplastics between field monitoring (i.e., exposure) and laboratory bioassays (i.e., effect), limiting the validity of ecological risk assessments of microplastics (Shim and Thompson, 2015). It is imperative to obtain exposure data for microplastics within a size range that is readily ingestible by

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aquatic organisms, while also obtaining effect data for larger-sized non-spherical microplastics that are technically detectable in real environments.

The particle toxicity of microplastics is affected by their size and shape. In general, the toxicity of non-spherical microplastics (i.e., fragments and fibers) is higher than that of spherical microplastics. For example, Ziajahromi et al. (2017) compared the toxicity of fiber and spherical microplastics using *Ceriodaphnia dubia* as a test species and concluded that the acute median lethal concentration and chronic median effect concentration of fiber microplastics were about 6 and 9 times lower than those of spherical microplastics, respectively. The toxicity also varies according to the size of the microplastic. Gray and Weinstein (2017) evaluated the effects of microplastic size on grass shrimp (*Palaemonetes pugio*) and found that microplastics larger than 50 μm had higher toxicity than microplastics smaller than 50 μm when the organisms were exposed to same microplastic density (i.e., 500,000 particles/L).

Microplastics larger than 300 μm are readily removed by conventional wastewater treatment systems and it is estimated that about 90% of the microplastics detected in the environment with current analytical methods are in the size range 20–300 μm (Song et al., 2014). In addition, because invertebrates favorably ingest <300 μm microplastics (Van Cauwenberghe et al., 2015; Van Cauwenberghe and Janssen, 2014), it is not likely that microplastics larger than 300 μm present a significant ecological risk to small aquatic organisms. Considering microplastic shapes, Ivar do Sul and Costa (2014) analyzed global environmental microplastic monitoring data at 35 locations and reported that fibers and fragments were mainly found at 31 locations, whilst spherical microplastics were only found at 4 locations. Therefore, considering the environmental distribution and toxicological characteristics of microplastics, it is necessary to specify their size and shape to better characterize the risk posed by microplastics in real aquatic environments.

Three papers on ecological risk assessment of microplastics have been published (Besseling et al., 2019; Burns and Boxall, 2018; Everaert et al., 2018). In those studies, the predicted or measured environmental concentrations (PECs/MECs) of microplastics were compared to their predicted no-effect concentrations (PNEC) or hazardous concentration for protecting 95% of the species (HC₅) to conclude that the PECs/MECs of microplastics were far lower than their PNEC/HC₅ values; therefore, no significant ecological concern was anticipated at this time. However, unfortunately, those earlier studies did not consider the ecologically relevant sizes or shapes of microplastics in their risk assessment. While Burns and Boxall (2018) derived the HC₅ value of microplastics in the size range 10–5000 μm , they did not consider environmentally relevant shapes. Everaert et al. (2018) and Besseling et al. (2019) derived the PNEC of microplastics based on all available toxicity data for microplastics without considering their sizes or shapes.

In the present study, we conducted an aquatic ecological risk assessment for non-spherical microplastics in the size range 20–300 μm in Korean marine environments covering coastal, continental shelf and deep-sea areas. Most toxicity studies on microplastics have focused on characterizing the toxicity of small (e.g., <131 μm) spherical microplastics (Burns and Boxall, 2018), presumably due to the ease of availability of those types of microplastics for toxicity testing, and a few toxicity studies on non-spherical microplastics have been conducted. However, recently, some researchers have begun to study the toxicity of non-spherical microplastics and some toxicity data (though not abundant) was available for fiber or fragment microplastics for our environmentally relevant ecological risk assessment (Imhof et al., 2017; Kalčíková et al., 2017; Martínez-Gómez et al., 2017; Qiao et al., 2019; Rochman et al., 2017; Weber et al., 2018; Ziajahromi et al.,

2017). The risk assessment in this study followed a generic framework (i.e., exposure assessment, ecological effects assessment, and risk characterization). To the best of our knowledge, this is the first study to consider both the size and shape of microplastics in both effect and exposure analyses for an ecological risk assessment.

2. Methods

2.1. Seawater sample collection

Microplastic abundance data in coastal seawater were obtained from a previous study (Song et al., 2018), and an additional survey was conducted in an enclosed bay to obtain seasonal abundance data. Seawater samples from the surface and the water column (mid-column and/or bottom water) were collected at 46 sampling stations in nine coastal areas including six bays (CS, HP, DR, GY, MS, and YI) off the coast of South Korea. Samples were collected in July or August of 2016 and 2017 for eight of the areas, with the exception of MS (Fig. 1); at MS, both surface and bottom water were collected at five stations in May, July, September and December 2018 to evaluate seasonal variation. A total of 163 water samples were collected from the nine coastal areas. Surface water samples with a volume of 100 L were manually collected from the top 20 cm using a customized surface water sampler made from a stainless tray, and subsurface water samples with a volume of 100 L were collected using a submersible pump (PD-272, Wilo; flow rate 140 L/min) after discarding the first 10 s of pumped water at each sampling depth (Song et al., 2018).

Surface water and a series of water column samples were collected at 14 stations in continental shelf areas of the Yellow Sea (average depth, 44 m; maximum depth 103 m) and the South Sea (between Jeju and Korea Straits; average depth, 100 m; maximum depth 210 m) of Korea in July and August 2018, and at 8 stations in deep sea areas of the East Sea (average depth, 1361; maximum depth 4049) in July and August 2019 (Fig. 1), from the R/V Eardo. At each station, four to six samples were taken vertically depending on the water depth. Surface water was collected 1 m below the water surface, and in the bottom layer, samples were collected 2–210 m above the sea floor, depending on the depth. A total of 102 water samples were collected from continental shelf and deep sea areas. Surface and subsurface water samples with a volume of 200 L were collected in the Yellow Sea and South Sea using a submersible pump after discarding the first 10 s of pumped water at each sampling depth. Bottom water samples were not available at three stations (YS7, SS4 and SS5) due to pump failure below 80 m depth. In the East Sea, surface water samples were collected using a submersible pump while deeper waters were collected using a rosette sampler with Niskin bottles due to their depth. All water samples from the coastal, shelf, and deep-sea areas were filtered through a 20 μm polyester mesh portable net to obtain reduced-volume samples. The filtered sample in the cod end of the net was transferred to a 1 L pre-cleaned amber glass bottle.

2.2. Microplastic analyses

The sample pretreatment and microplastic identification methods have been described in detail elsewhere (Song et al., 2018; Eo et al., 2019). In a previous study, we separated microplastics with saturated NaCl solution (1.2 g/cm³) (Song et al., 2018), however, it was later found that high density microplastics over 1.2 g/cm³ were present in subsurface and even in surface water samples. Hence, the density separating solution was changed from saturated NaCl to lithium metatungstate (LMT, 1.6 g/cm³) solution (Eo et al., 2019).

Each sample was passed through a 20 μm metal sieve (10 cm Ø)

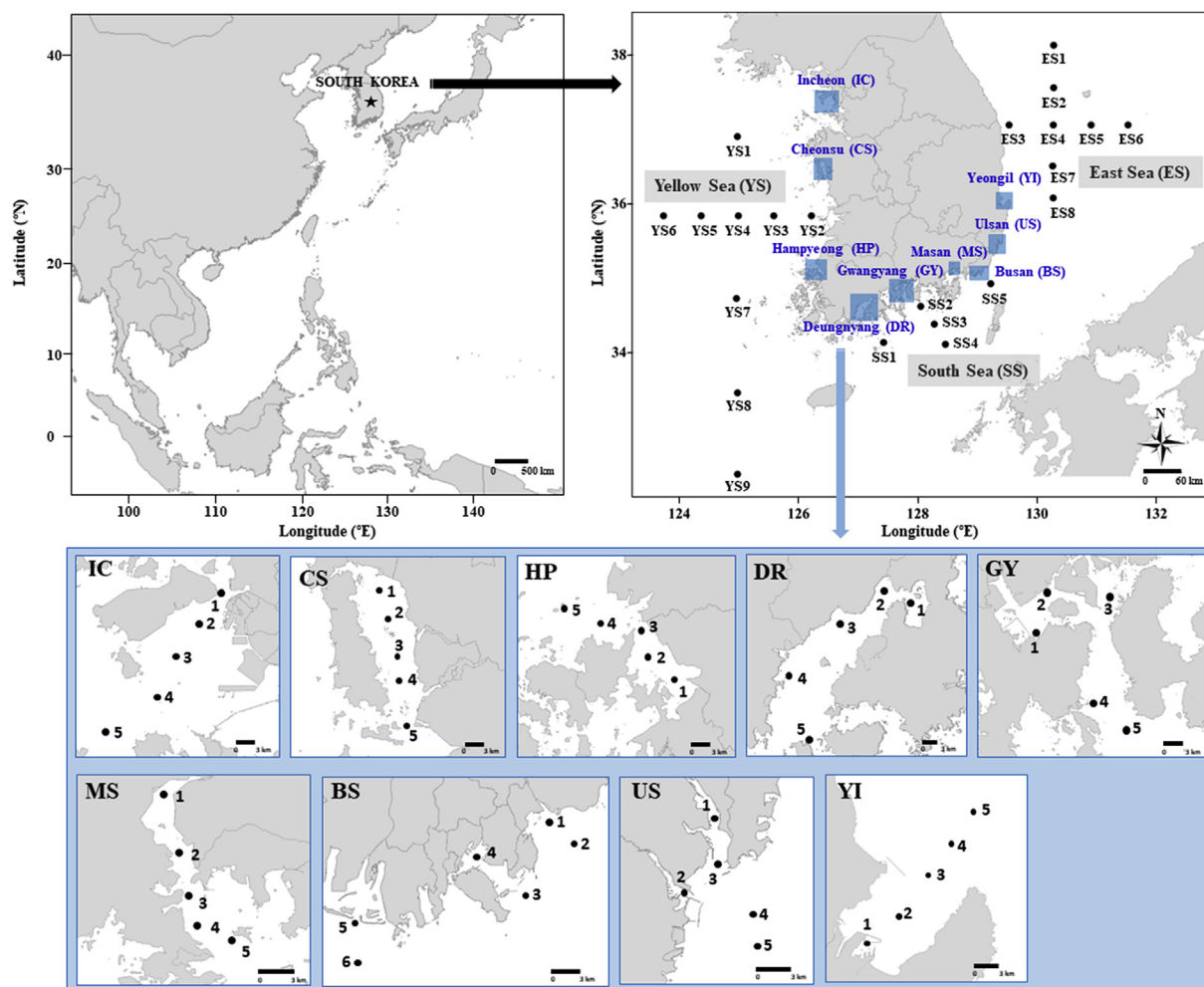


Fig. 1. Location map of the surface water and water column microplastic sampling stations in three continental shelf and deep sea areas (Yellow Sea, South Sea and East Sea) and nine coastal areas (blue boxes; IC, CS, HP, DR, GY, MS, BS, US, and YI) of South Korea. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and washed with high-performance liquid chromatography (HPLC) grade water, leaving only solid particles. Particles remaining on the sieve were transferred to a 250 mL glass beaker using a squeeze bottle containing HPLC-grade water, and then dried at 60 °C in a drying oven until all water had evaporated. After drying, the organic matter in the beaker was digested using 20 mL 35% H_2O_2 and Fe (II) solution on a hot plate (60 °C, 180 rpm). The resulting mixture was passed through a 20 μm sieve to remove the remaining solution and then washed with HPLC-grade water. The particles were transferred to a 120 mL glass density separation funnel using LMT (2.95 g/cm^3) solution diluted to 1.6 g/cm^3 and allowed to settle overnight. The settled particles were drained, and the remaining solution was filtered using 5 μm filter paper (polycarbonate membrane filter; 47 mm $\varnothing$; Millipore). All particles that appeared to be plastic on the filter were identified using a Fourier transform infrared (FT-IR) microscope (Nicolet iN10 or Nicolet 6700 and Continuum; Thermo Fisher Scientific, USA) in attenuated total reflectance (ATR) mode. Microplastic particles were randomly examined over 12.5–100% of the total filter area. Whole plastic-like particles on the filter paper were selected under a microscope (100 or 150 $\times$), and their polymer type was immediately identified using FT-IR. The spectra were recorded as the average of 16–128 scans in the spectral range of 650–400 cm^{-1} at a resolution of 8 cm^{-1} .

A blank test was conducted using HPLC-grade water poured in

the field to check for contamination during the experimental process. Blank samples ($n = 34$) were collected, and subjected to the same analytical procedures as those used for sample analyses. All plastic-like particles on the filter were identified using an FT-IR microscope and 0–2 particles were found in each blank sample. A value of three times the mean particle number (0.9 particles/filter) identified from the field blank samples was set as the detection limit (2.7 particles/filter). All microplastic numbers counted on filters from samples, except for five samples (in the range of 3–5 particles/filter), were well above the detection limit. No plastic particles were detected on a wet filter paper exposed to air in the laboratory during the FT-IR observation time (2–5 h).

At each step, containers were covered with aluminum foil to avoid contamination from the air. All solutions except HPLC-grade water were filtered using GF/F filters (0.7 μm , 47 mm $\varnothing$), and all equipment was rinsed at least three times with HPLC-grade water prior to use. One 20 μm sieve was used for each sample, and the sieve was rinsed and shot from behind with an air gun to remove particles trapped in the sieve prior to use. Cotton lab gowns and latex gloves were worn at all times.

2.3. Microplastic exposure data

In the microplastic analytical procedure, microplastic particles

were categorized by shape (fragment, fiber, and sphere) and size to within 50 μm intervals ranging from 20 μm to 5 mm. From these, only non-spherical type (fragment and fiber) microplastic particles in the size range 20–300 μm were extracted and used for further risk assessment, to match with the toxicity data.

2.4. Microplastic risk assessment

The risk assessment for microplastics in seawater was conducted by comparing the detected concentrations at each seawater sampling site to the derived PNEC. If the detected concentrations exceeded the PNEC value for a site, it was determined that the site warrants further risk characterization. By contrast, if the detected concentrations were lower than the PNEC value for a site, we concluded that the microplastic concentrations at the site do not pose any ecotoxicological threats to the ecosystem, and therefore the site does not need further characterization.

The PNEC values for fragment and fiber microplastics in the size range 20–300 μm were derived using a species sensitivity distribution (SSD) approach. First, we collected 72 chronic toxicity data points (e.g., no observed effect concentration [NOEC], lowest observed effect concentration [LOEC] and 10% effective concentration [EC₁₀]) for microplastics with ecologically relevant endpoints (i.e., mortality, growth, development, reproduction and population growth) regardless of the microplastic shape and size (see Table S1 and List S1 of the Supplementary Materials). Then the toxicity data were screened for fragment and fiber microplastics in the size range of 20–300 μm .

To supplement the toxicity data, we also conducted a 21-day chronic-toxicity exposure test with 100 μm fiber microplastics on fish species (*Cyprinodon variegatus*) and observed no mortality at a microplastic concentration of 5 mg/L, which corresponds to about 1.33×10^5 particles/L. Then this experimental toxicity data was used to assess the risk of microplastic contamination. Details of the toxicity testing conditions can be found in Table S2. Finally, a total of 11 chronic toxicity data points covering eight taxonomic classes (i.e., fish, crustacean, algae, echinoderm, mollusk, macrophyte, coral, and amphipod) were screened and used for SSD derivation (Table 1). If multiple toxicity values were available for a species, the EC₁₀ or NOEC value was used for SSD derivation; however, if neither of these was available for a species, the LOEC value for the species was converted into the NOEC by dividing the LOEC by 2, and then this was used as the representative toxicity value (ECHA, 2008).

For the case when the toxicity value was expressed as a unit of mass per volume, the unit was converted into particle number per volume using the following equations with the assumption that fragments and fibers are spherical and cylindrical, respectively:

For fragments:

$$\text{particle abundance} = \frac{\text{mass concentration}}{d \cdot \frac{4}{3} \pi r^3}$$

For fibers:

$$\text{particle abundance} = \frac{\text{mass concentration}}{d \cdot \pi r^2 \cdot l}$$

where *particle abundance* and *mass concentration* are the microplastic concentrations expressed in units of particle number/volume and mass/volume, respectively; *d* refers to the particle density of microplastics; and *r* and *l* are the radius of fragment and fiber microplastics and the length of fiber microplastics, respectively. All toxicity data from the literature provided information about the radius of fragment microplastics and length of fiber microplastics. When there was no information about the radius of fiber microplastics, we assumed that the diameter of the fiber microplastics was 20 μm , which is the median diameter of fiber microplastics observed in the environment (i.e., size range 10–30 μm) (Cole et al., 2014). If density information was not given in the toxicity data literature, average density values with respect to the type of microplastic were applied (Duis and Coors, 2016).

The SSD approach was developed based on the collected chronic toxicity values using the SSD Generator developed by the United States Environmental Protection Agency (USEPA) (USEPA, 2014). The SSD Generator fits the toxicity data to a linearized lognormal distribution. Our goodness-of-fit analyses revealed that the chronic toxicity values for SSD derivation followed a lognormal distribution; therefore, the use of the SSD Generator did not cause statistical bias. The chronic toxicity data on marine and freshwater species were combined in SSD development to enhance their statistical power. The HC₅ value was determined based on the SSD and then the PNEC value was calculated by dividing the HC₅ by an assessment factor of 5. The toxicity data met the minimum quantitative requirements for PNEC derivation using an SSD approach (i.e., ≥ 10 chronic toxicity data covering ≥ 8 taxonomic classes). However, we assigned the assessment factor as the highest value (i.e., 5) because chronic toxicity data on insects such as midges were unavailable (a qualitative requisite for PNEC derivation using an SSD approach) (ECHA, 2008), and freshwater and marine toxicity data were combined for PNEC derivation.

2.5. Statistical analyses

Statistical analyses were performed using the SPSS 19.0 software (IBM Corp., Armonk, NY, USA). Seasonal and vertical distribution comparisons of microplastic concentrations in seawater were conducted using the Mann-Whitney U test, the Kruskal–Wallis test or the one-way ANOVA test. Results with $p < 0.05$ were considered statistically different.

Table 1

The chronic no observed effect concentrations (NOECs) used for the derivation of predicted no effect concentration (PNEC) of non-spherical (fragment and fiber) microplastics in the size range of 20–300 μm .

Species	Environment	Taxonomy	Size (μm)	NOECs (particles/L)	Exposure duration (days)	Endpoint	Source
<i>Ceriodaphnia dubia</i>	freshwater	crustacean	100–400	2.40E+03	8	Reproduction	Ziajahromi et al. (2017)
<i>Cyprinodon variegatus</i>	marine	fish	12–704	1.33E+05	28	Mortality	this study
<i>Danio rerio</i>	freshwater	fish	<250	3.40E+02	21	Growth	Qiao et al. (2019)
<i>Daphnia magna</i>	freshwater	crustacean	29.5 $\pm$ 26	5.80E+05	20–22	Growth	Imhof et al. (2017)
<i>Gammarus fossarum</i>	freshwater	amphipod	32–250	1.67E+05	28	Growth	Straub et al. (2017)
<i>Gammarus pulex</i>	freshwater	crustacean	10–150	4.00E+06	48	Mortality	Weber et al. (2018)
<i>Heliopora coerulea</i>	marine	coral	65–410	1.00E+02	168	Growth	Reichert et al. (2019)
<i>Lemna minor</i>	freshwater	macrophyte	40	1.19E+04	7	Growth	Kalčíková et al. (2017)
<i>Microcystis arrginosa</i>	freshwater	algae	<200	1.20E+06	21	Growth	Yokota et al. (2017)
<i>Oyster edulis</i>	marine	mollusk	0.48–316	2.11E+03	60	Growth	Green (2016)
<i>Paracentrotus lividus</i>	marine	echinoderm	<80	3.87E+03	2	Development	Martínez-Gómez et al. (2017)

3. Results and discussion

3.1. Microplastic exposure levels in coastal water

Non-spherical microplastics in the size range 20–300 μm were detected in all regions and stations except for a mid-column sample at St.4 at CS (Table 2, S3, S4). Spherical microplastics were detected at 16 out of 46 surveyed stations in coastal areas. They only accounted for 1% of the total microplastic abundance in the study area (80% of fragments and 19% of fiber). This indicates that ecological risk assessment that uses bioassay data obtained with spherical type microplastics may be more uncertain, because of the mismatch in the target microplastic. The abundance of each sample ranged from below the detection limit at CS to 4.28 particles/L at IC. The highest mean and median abundances of 1.57 and 0.84 particles/L were found at IC. The other coastal regions exhibited a narrow average abundance range, from 0.29 to 0.79 particles/L (median range 0.14–0.50 particles/L). The mean abundance of non-spherical microplastics in the size range 20–300 μm (1.14 ± 0.88 particles/L) in surface water samples from the nine coastal areas of Korea ($n = 61$) sampled in this study (Table S5) was 407 times higher than the average abundance of microplastics including spherical type microplastics in the size range 300–5000 μm (0.00285 particles/L) collected using a Manta trawl in eight coastal areas of Korea in 2015–2017 ($n = 83$) (Kwon et al., 2020). This difference of over two orders of magnitude in the microplastic abundances between the two size ranges 20–300 μm and 300–5000 μm in real environments indicates that ecological risk assessments should consider the size range of microplastics in both exposure and effect data.

Seasonal variation in the microplastic abundance in surface and bottom waters were evaluated at Masan Bay. The abundance of microplastics in surface seawater was the highest in July (1.28 ± 0.82 particles/L) after the Asian monsoon, followed by December (0.84 ± 0.57 particles/L), May (0.62 ± 0.27 particles/L), and September (0.36 ± 0.10 particles/L), however, the difference was not statistically significant (one-way ANOVA; $p > 0.05$) (Table S4). The abundance of microplastics in bottom seawater was highest in May (0.28 ± 0.12 particles/L), followed by July (0.20 ± 0.16 particles/L), December (0.20 ± 0.05 particles/L), and September (0.17 ± 0.09 particles/L), but the difference was not significant (Kruskal-Wallis test, $p > 0.05$). Although the adjacent freshwater Nakdong River showed a significant seasonal variation in microplastic abundance (Eo et al., 2019), there were no significant differences between the wet (July) and the dry (February) season in the surface and bottom water of Masan Bay. However, the highest microplastic abundance (2.64 particles/L) was detected in

surface water from the innermost station in July, which indicates that river discharge may locally influence Masan Bay during the wet season.

The mean abundance of non-spherical microplastics in surface water (1.14 ± 0.88 particles/L) was 3.7 times (Mann-Whitney U test, $p < 0.05$) higher than the mid-column and bottom water (0.31 ± 0.32 particles/L) in coastal regions (Table S5). Coastal water sampling using a small boat allowed manual collection of surface water including the sea surface microlayer, where a high abundance of microplastics accumulates (Song et al., 2014). Although microplastic abundances were relatively low in the mid-column and bottom water compared to the surface water, including the sea surface microlayer, many aquatic organisms will uptake microplastics in the subsurface waters in which they dwell. Therefore, when only surface water microplastic data are used for ecological risk assessment, real exposure levels could be overestimated.

Fragment type microplastics exhibited a peak abundance in the size range 100–150 μm , while fiber type microplastics exhibited a peak in the size range 200–250 μm . Among the 24 synthetic polymers identified with FT-IR spectroscopy, polypropylene, polyethylene, polyacrylate (sum of polybutyl methacrylate and acrylic), and polyethylene vinylacetate accounted for 35, 18, 12, and 11% of the total abundance, respectively.

3.2. Microplastic exposure levels in continental shelf and deep sea water

Non-spherical microplastics in the size range 20–300 μm were detected at all stations in the continental shelf and deep sea areas of Korea (Table 2, Table S6). Spherical microplastics were only detected in 7 surface and (or) subsurface samples at 5 out of the 44 surveyed stations. Spherical microplastics only accounted for 0.4% of the total microplastic abundance in the study area. The mean $\pm$ standard deviation (range of) abundances of non-spherical microplastics (20–300 μm) were 0.19 ± 0.31 (0.01–1.68) particles/L for the Yellow Sea, 0.09 ± 0.09 (0.01–0.41) particles/L for the South Sea, and 0.52 ± 1.31 (0.03–7.88) particles/L for the East Sea. The highest microplastic abundance of 7.88 particles/L was found in the bottom water (sampling depth, 2100 m) at ES6 in the East Sea.

The mean abundances of microplastics in the surface (V1–2), thermocline (V3–4) and bottom (or deep) layers (V5–6) were 266 ± 459 , 228 ± 330 , and 136 ± 104 particles/ m^3 in the YS and SS, and 289 ± 280 , 260 ± 125 , and 1136 ± 2043 particles/ m^3 in the ES, respectively (Table 2; Table S6). There were no significant differences in the mean abundances of microplastics in the three vertical

Table 2

The summary of non-spherical type microplastic (20–300 μm) abundance in surface and sub-surface seawater from coastal, continental shelf, and deep sea area of South Korea.

Region		# of data	Abundance (particles/L)				
			Min	Max	Mean	SD	Median
Coast	Incheon (IC)	15	0.22	4.28	1.57	1.33	0.84
	Cheonsu (CS)	15	ND ^a	0.62	0.29	0.21	0.26
	Hampyeong (HP)	15	0.04	1.34	0.38	0.42	0.14
	Deungnyang (DR)	15	0.06	1.18	0.43	0.37	0.22
	Gwangyang (GY)	15	0.11	2.36	0.79	0.76	0.50
	Masan (MS)	40	0.06	2.64	0.50	0.51	0.36
	Busan (BS)	18	0.02	1.18	0.44	0.39	0.28
	Ulsan (US)	15	0.06	3.08	0.67	0.82	0.26
	Yeongil (YI)	15	0.08	2.24	0.78	0.60	0.60
	Yellow Sea (YS)	41	0.01	1.68	0.19	0.31	0.10
Continental shelf	South Sea (SS)	22	0.01	0.41	0.09	0.09	0.07
	East Sea (ES)	39	0.03	7.88	0.52	1.31	0.15

^aND: Not detected.

layers in any of the areas (Kruskal-Wallis test, $p > 0.05$) due to the high variability between stations.

Following the same size distribution trend as for coastal waters, fragment and fiber type microplastics in continental shelf and deep water areas exhibited peak abundances in the size ranges 100–150 μm and 200–250 μm , respectively. Among the 24 synthetic polymers identified through the FT-IR spectroscopy, polypropylene (44%) was the most abundant polymer, followed by alkylid (23%), and polyacrylate (sum of polyacrylate copolymers and polymethyl methacrylate) (10%).

Overall, the mean and median abundances were observed in the order of coastal, deep sea, and continental shelf areas (Fig. 2, Table S7). Although the highest (7.88 particles/L) and relatively high microplastic abundances (2.54 and 2.19 particles/L) were detected in deep sea water samples (1200–2100 m) from the East Sea, coastal waters exhibited relatively high abundances compared to continental shelf and deep sea environments. These results indicate that aquatic organisms could be exposed to relatively high microplastic levels in coastal areas compared to open sea areas, except for a few deep sea hot spots.

3.3. PNEC derivation

3.3.1. Chronic toxicity data analyses

The cumulative distributions of the 72 chronic toxicity data points collected in this study were plotted according to microplastic shape and size, and are shown in Fig. 3. The cumulative distribution for non-spherical microplastics with a size $\geq 20 \mu\text{m}$ is plotted on the far left side of the graph. This indicates that non-spherical microplastics $\geq 20 \mu\text{m}$ may pose greater ecotoxicity to aquatic organisms on a number basis, compared to other types of microplastics (i.e., spherical and/or small [$<20 \mu\text{m}$] microplastics). Previous toxicological studies have also confirmed that non-spherical and large microplastics are more toxic to aquatic organisms than spherical and/or small ones (Gray and Weinstein, 2017; Ziajahromi et al., 2017). Moreover, because large ($>20 \mu\text{m}$) and non-spherical microplastics are dominantly detected in real-world environments, it is more practical and environmentally relevant to assess the risk posed by large and non-spherical microplastics rather than including all types of microplastics.

3.3.2. PNEC derivation

The developed SSD based on the 11 chronic toxicity data points covering 8 taxonomic classes (i.e., fish, crustacean, algae, echinoderm, mollusk, macrophyte, coral and amphipod) for fragment and

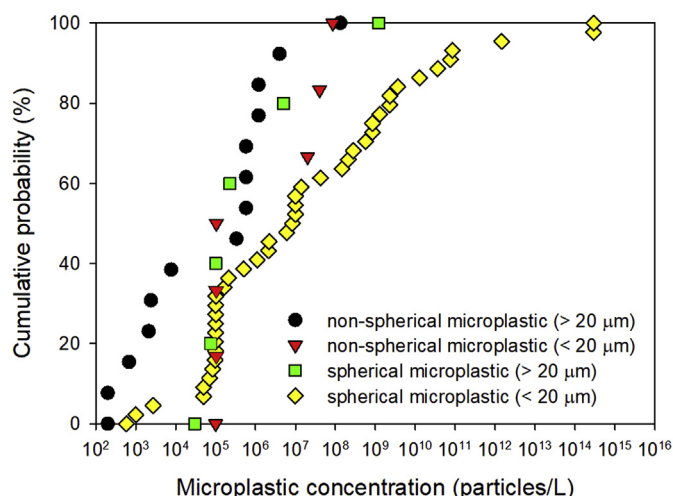


Fig. 3. Cumulative distributions of chronic toxicity values (no observed effect concentration (NOEC), lowest observed effect concentration (LOEC), median effect concentration (EC_{50}) and 10% effect concentration (EC_{10})) of microplastics collected in this study according to size and shape. Solid circle = non-spherical microplastic ($\geq 20 \mu\text{m}$); open circle = non-spherical microplastic ($< 20 \mu\text{m}$); solid triangle = spherical microplastic ($\geq 20 \mu\text{m}$), and open triangle = spherical microplastic ($< 20 \mu\text{m}$). Endpoints included lethality, growth, reproduction, development and population growth.

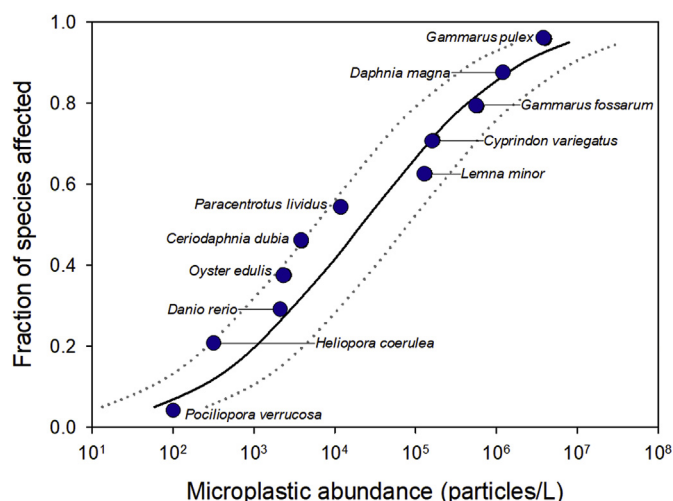


Fig. 4. Species sensitivity distribution (SSD; black line) plotted with 95% confidence interval (CI; gray line) based on chronic no observed effect concentrations (NOECs) of non-spherical microplastics in the size range of 20–300 μm . Endpoints included lethality, growth, reproduction, development and population growth. The information on the toxicity data depicted in this figure (e.g., taxonomy of the tested organisms, microplastic size, NOEC, exposure duration and endpoint) are listed in Table 1.

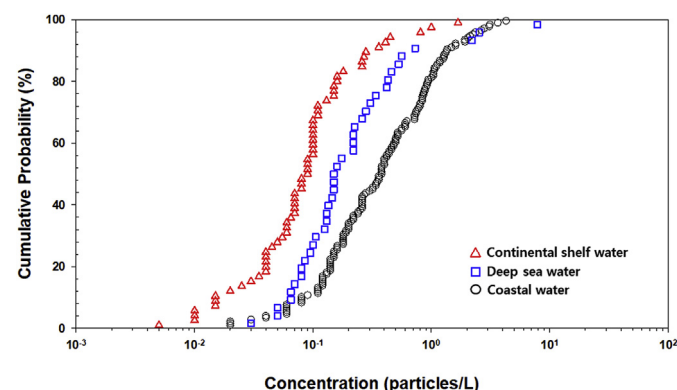


Fig. 2. Comparison of non-spherical (fragment and fiber) microplastic (20–300 μm) abundance in surface and sub-surface seawater from coastal (nine regions; $n = 163$), continental shelf (Yellow Sea and South Sea of Korea; $n = 63$), and deep sea (East Sea; $n = 39$) areas of South Korea.

fiber microplastics in the size range 20–300 μm is shown in Fig. 4, while the HC_5 and PNEC values derived based on the SSD are listed in Table 3. The derived HC_5 was 59.8 particles/L (95% confidence interval [CI]: 13.7–261.6 particles/L), and then the PNEC was estimated to be 12.0 particles/L (95% CI: 2.7–52.3 particles/L) by dividing the derived HC_5 value by an assessment factor (AF) of 5. This is because chronic toxicity data on insects such as midges, a qualitative requisite for PNEC derivation using an SSD approach, were unavailable (ECHA, 2008).

For comparison, we also derived the SSD for all types of microplastics regardless of size and shape based on the chronic NOEC toxicity data for 33 aquatic species (Fig. 5 and Table S8). Both SSDs overlapped, but the span of confidence intervals for the SSDs

Table 3

The 5% hazardous concentration (HC₅) and predicted no effect concentration (PNEC) of fragment and fiber type microplastics in the size range of 20–300 µm expressed with 95% confidence interval (CI) derived from species sensitivity distribution (Fig. 4).

Parameter	Value (particles/L)		
	Central tendency	95% upper CI	95% lower CI
HC ₅	59.8	261.6	13.7
PNEC (=HC ₅ /5)	12.0	52.3	2.7

were different. When the toxicity data for all types of microplastic regardless of shape and size were included for SSD derivation, the confidence interval of the SSD spanned a factor of 52. On the other hand, when only toxicity data for non-spherical microplastics in the size range 20–300 µm were included, the span of the confidence interval decreased to a factor of 19, although limited toxicity data were used for SSD derivation. Because both SSDs had similar r^2 values (i.e., 0.966 and 0.960), the decrease in the span of the confidence interval is mainly due to a reduction in the span of the toxicity datasets, as shown in Fig. 5. Moreover, the spans of the confidence intervals of the SSDs reported in previous studies (Besseling et al., 2019; Burns and Boxall, 2018; Everaert et al., 2018) were about 100, which is about 5 times larger than the span value derived in this study. This indicates that ignoring the shape and/or size of microplastics can reduce the statistical reliability when deriving a PNEC value, even if more toxicity data are available for SSD derivation. Moreover, because spherical microplastics were only rarely detected in the sampling sites in this study, and because small microplastics cannot be detected with current technologies, we believe that it is more environmentally relevant and reliable to exclude spherical and/or small microplastics (<20 µm) when deriving the PNEC.

3.4. Risk characterization

We conducted a risk characterization of fragment and fiber microplastics in the size range 20–300 µm by comparing the MECs in coastal, continental shelf and deep sea environments to the derived PNEC (i.e., 12 particles/L) value. If all MECs are lower than the PNEC for a site, then it was concluded that no risk is anticipated for that site and no further assessment was necessary. By contrast, if any MECs exceeded the PNEC for a site, it was determined that the organisms at the site could be at risk for microplastic toxicity.

The results of the comparisons between the MEC and PNEC values revealed that the MECs detected in all marine environments in South Korea did not exceed the derived PNEC value (Fig. 6). The highest microplastic level was 7.88 particles/L, about two-thirds of the derived PNEC in this study (Tables 2 and 3). It can therefore be concluded that the current microplastic pollution levels in Korean marine environments do not have adverse effects on the aquatic ecosystem.

Our risk assessment results are consistent with earlier ones (i.e., no ecological risk is anticipated from microplastics) (Besseling et al., 2019; Burns and Boxall, 2018). These studies suggest that the MEC of microplastics (8.5–100 particles/L) is far lower than the HC₅ value of 1016–64,000 particles/L, derived using an SSD approach based on 9–10 chronic toxicity data points for microplastics. Earlier studies did not consider the shape of microplastics in aquatic ecological risk assessment, however there was a large variation between the HC₅ values derived in earlier studies and in this study. The derived HC₅ value in this study (58.7 particles/L) was two or three orders of magnitude lower than the values derived in earlier studies (1016–64,000 particles/L). Because this study and

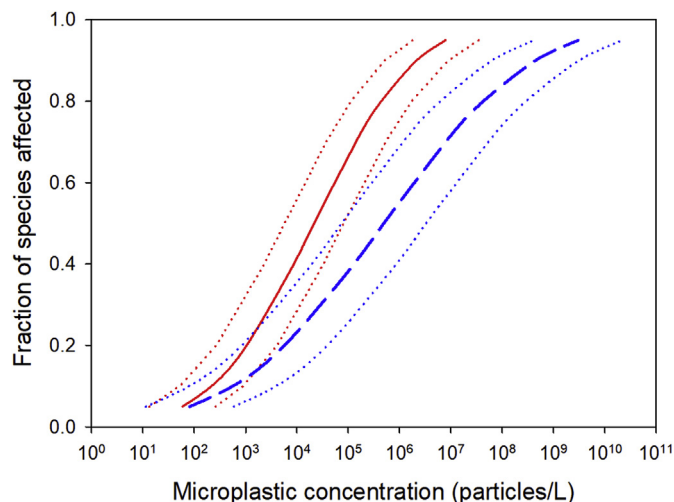


Fig. 5. Comparison between the species sensitivity distribution (SSD) based on chronic toxicity data for microplastics regardless of shape and size, and non-spherical microplastics in the size range of 20–300 µm. Blue (broken line) and red (straight line) and red are the SSD for microplastics regardless of shape and size, and non-spherical microplastics in the size range of 20–300 µm, respectively. Blue and red dotted lines are the confidence intervals of the SSD for microplastics regardless of shape and size, and non-spherical microplastics in the size range of 20–300 µm, respectively. End-points included lethality, growth, reproduction, development and population growth. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

earlier studies employed the same SSD Generator developed by the USEPA, the variation in the HC₅ values does not arise from the statistical method used. Instead, this variation was mainly due to the difference in the toxicity values used for HC₅ derivation. The range of toxicity values used for HC₅ derivation in this study (340–4,000,000 particles/L) was lower than those in Burns and Boxall (2018) and Besseling et al. (2019) (100,000–10,000,000, and 1000–100,000,000 particles/L, respectively). Moreover, different toxicity values were used for the same species in these studies. For example, Burns and Boxall (2018) suggested *Danio rerio* as the most-sensitive species and set the toxicity value for *D. rerio* at about 100,000 particles/L. This toxicity value was based on toxicity test results from 45 µm spherical microplastics (Chen et al., 2017). In this study, *D. rerio* was also considered a highly sensitive species, but the toxicity value for *D. rerio* was set at 340 particles/L based on the chronic toxicity test results for fiber microplastics smaller than 250 µm. This indicates that inclusion of toxicity data for spherical microplastics that are only rarely detected in the environment and less toxic than non-spherical microplastics may lead to significant underestimation of the actual risk posed by microplastics.

4. Future perspective

Our risk assessment identified that the current status of microplastic pollution does not seem to pose significant risks to the aquatic ecosystem. However, this does not necessarily mean that the aquatic ecosystem is not at risk for microplastic toxicity in the future, because of the dramatically increasing trend of microplastic concentrations in the aquatic environment.

The exponential increase in the abundance of microplastics in marine environments over the past several decades has been evidenced by sediment core (Brandon et al., 2019) and time series data of plastic debris entanglement from continuous plankton recorder sampling (Ostle et al., 2019). If current world plastic production rates and plastic waste management policies are maintained, it is predicted that a fourfold increase in floating microplastic

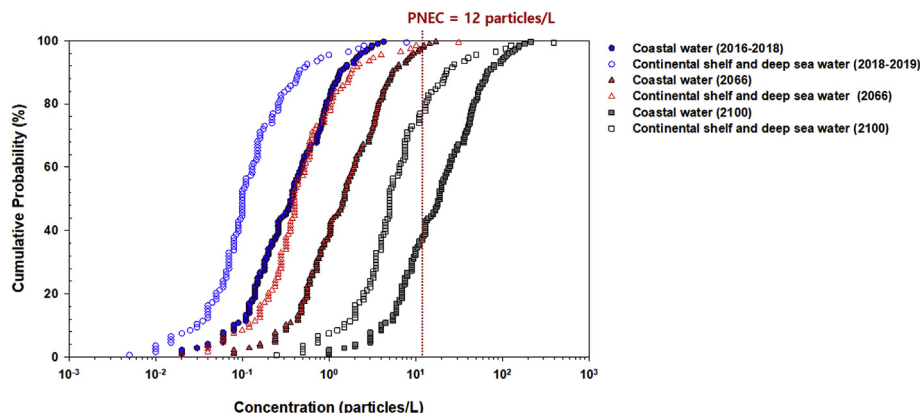


Fig. 6. Present and future expected (years of 2066 and 2100) environmental abundance of fragment and fiber microplastics in the size range of 20–300 μm in surface and water column of coastal areas and continental shelf and deep sea areas (Yellow Sea, South Sea and East Sea) in Korea. The circles are current exposure levels, and the triangles and squares are predicted exposure levels in 2066 and 2100, respectively. The predicted no effect concentration (PNEC) of 12 particles/L was derived in this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

abundance might be observed in the North Pacific by 2066 (Isobe et al., 2019) and a 50-fold increase by 2100 in world oceans (Everaert et al., 2018), compared to current concentrations. When these predictions are applied to future microplastic levels in Korean marine environments, 0.6% of the continental shelf and deep sea and 2.2% of the predicted coastal microplastic abundances will exceed the PNEC value by 2066, and 22.1% of the continental shelf and deep sea and 61.6% of the coastal microplastic abundances will exceed the PNEC value in 2100 (Fig. 6). This implies that microplastic pollution will have an ecological impact on both coastal and open sea environments in the future and thus proper precautionary actions to prevent an increase in microplastic concentrations in the aquatic environment are warranted.

5. Conclusions

A mismatch in the sizes and shapes of microplastics analyzed in field monitoring and laboratory bioassays has made ecological risk assessment difficult (Shim and Thompson et al., 2015). To overcome this mismatch, we analyzed microplastics with sizes down to 20 μm for the purpose of ecological risk assessment. Target microplastics in the size range 20–300 μm were included in the risk assessment, as they were predominantly observed at our study sites and showed adverse biological effects at lower exposure levels on a particle number basis than other types of microplastics (e.g., spherical and/or small microplastics). Derivation of the PNEC for microplastics in marine environments was conducted using an SSD approach and non-spherical microplastics in the size range 20–300 μm to match with the available exposure data. The observed non-spherical microplastic abundances in surface and subsurface seawater from coastal, continental shelf, and deep sea areas of South Korea were below the derived PNEC of 12 particles/L. However, the predicted microplastic concentrations at the current rate may be above the PNEC values in both coastal and open sea environments in the future. It is necessary to take preventive action to reduce or at least maintain current microplastic pollution levels in Korean marine environments, to protect such environments in the future.

Author contributions

J.W.J., J.W.P. and W.J.S conceptualized this study. J.W.J. carried out the risk assessment data analysis and writing-original draft under the supervision of J.W.P. and W.J.S. (writing-review &

editing). J.C performed toxicity data collection and visualization. S.E, Y.K.S., and Y.C. contributed to produce the microplastic monitoring data analyzed in this study and writing-original draft. S.H.H contributed to the project administration and funding acquisition. All authors contributed to the critical review of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2020.116217>.

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